10.6 Circumference and arc length

EASY

Learning intention We are learning to find circumferences, arc lengths and perimeters of curved shapes.

Success criteria
- ☐ I can use C = πd or C = 2πr to find a circumference
- ☐ I can find an arc length: $\frac{\theta}{360} \times 2\pi r$
- ☐ I can find the perimeter of sectors and composite shapes

NOTES — Key ideas for this lesson

TEACHER COPY

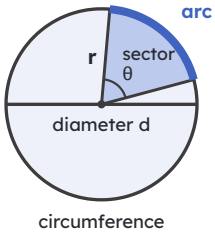
KEY TERMS

circumference (C): the distance around a circle
radius (r): from the centre to the edge
diameter (d): across through the centre; d = 2r
arc: part of the circumference
sector: a slice of a circle between two radii
θ: the angle at the centre of a sector
perimeter: the distance around the outside of a shape

FORMULAS

C = πd = 2πr
Arc length = $\frac{\theta}{360} \times 2\pi r$
Semicircle: P = ½πd + d
Quadrant: P = ¼ × 2πr + 2r
Sector: P = arc + 2r

STEPS 1 r = d ÷ 2 2 write the formula 3 substitute (π key) 4 round to 1 d.p. + unit



circumference

Level 1 · Getting started

Easy · 9 questions

WE DO With your teacher: copy what they write on the board.

Example 1 Find each circumference or perimeter, correct to one decimal place.

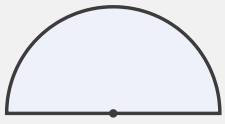
a Circumference



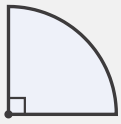
b Circumference



c Perimeter



d Perimeter



YOU DO On your own: **Easy** work down the box, one round at a time. Show your working. 9 QUESTIONS

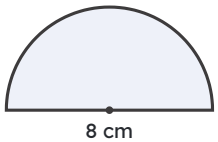
REMEMBER C = πd = 2πr · Semicircle P = ½ × πd + d · Quadrant P = ¼ × 2πr + r + r · Use the π key.

ROUND 1 Questions 1–6: find the circumference of each circle, correct to one decimal place. Is it r or d?

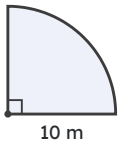
1. r = 3 cm
2. r = 7 m
3. r = 2.5 m
4. d = 8 cm
5. d = 15 m
6. d = 1.4 km

ROUND 2 Questions 7–9: find the perimeter of each shape, correct to one decimal place.

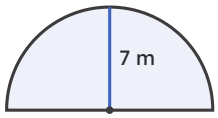
7.



8.



9.



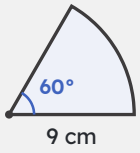
Level 2 · Building up

Medium · 9 questions

WE DO With your teacher: copy what they write on the board.

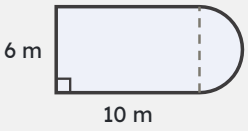
Example 2 **Give each answer correct to one decimal place.**

a Arc length



b Perimeter of the sector in part a

c Perimeter



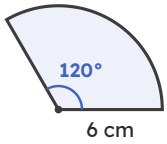
d Radius of a circle with circumference 50 cm

YOU DO On your own: **Medium** work in order. A tag shows an earlier lesson. Show your working.

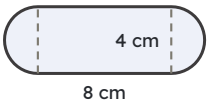
9 QUESTIONS

REMEMBER Arc = θ ÷ 360 × 2πr (θ = angle at the centre) · Sector P = arc + 2r · Composite: add the outside edges only

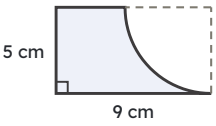
1 Perimeter



2 Perimeter



3 Perimeter. Cut-out: r = 5 cm



4 A kite has diagonals of 6 m and 5 m. Find its area. [10.5]

5 Find the arc length of a 120° sector with r = 3 m.

6 Write the limits of accuracy of 4.5 m. [10.2]

7 A circle has a circumference of 18.85 m. Find its radius.

8 Find the perimeter of a semicircle with d = 20 cm.

9 Percentage error of 20 cm, to the nearest cm? [10.3]

WORDED PROBLEMS Draw a diagram if it helps, then solve. Show your working.

3 PROBLEMS

1 A round garden bed has a diameter of 3.2 m. Edging costs $12 per metre. Find the cost to edge the bed, to the nearest dollar.

Answer

2 A bike wheel has a radius of 33 cm. How many full turns does it make to travel 1 km?

Answer

3 A running track is a rectangle 84 m long with a semicircle on each end. Each semicircle has a diameter of 64 m. Find one lap, to the nearest metre.

Answer